

UNIT - 1

2 MARKS QUESTIONS

1. Define AI.

Artificial Intelligence (AI) is the branch of computer science dedicated to creating systems and machines capable of performing tasks that typically require human intelligence. This includes activities such as learning, reasoning, problem-solving, perception, language understanding, and decision-making.

2. Define AI with respect to "Thinking humanly".

Artificial intelligence (AI) with respect to "Thinking humanly" is the branch of computer science and engineering that focuses on creating systems capable of creating task. It mimic how humans think, reason and solve problems. To achieve this, it uses the methods like cognitive science brings together computer models from AI and experimental techniques from psychology to try to testable theories of the working of human mind.

3. Define AI with respect to "Thinking rationally".

Artificial Intelligence (AI), with respect to thinking rationally, is the branch of computer science and engineering that focuses on creating systems that can reason, draw inferences, and make decisions in a logically consistent manner, similar to human rational thinking.

4. Define AI with respect to "Acting humanly".

Artificial Intelligence (AI), with respect to "Acting humanly," is the branch of computer science and engineering that focuses on creating systems capable of performing tasks and exhibiting behaviors indistinguishable from those of humans. This involves mimicking human actions, interactions, and responses to provide a seamless, human-like experience in various contexts.

5. Define AI with respect to "Acting rationally".

Defining AI with respect to "Acting Rationally" refers to the rational agent approach, which involves an agent that senses and acts. In this context, AI is about capturing intelligence abstractly, focusing on rationality rather than modelling human behaviour. According to Russell and Norvig's textbook "Artificial Intelligence: A Modern Approach", AI can be classified into four categories, with "Acting Rationally" being one of them. This approach deals with behaviour, where a system is considered rational if it does the "right thing" given what it knows, unlike humans who can make mistakes.

6. What is Turing Test?

Turing test are there to allow rational actions. thus, we need the ability to represent knowledge and reason with it because this enables us to reach good decisions in a wide variety of

situations. This test is significant in the field of artificial intelligence as it aims to assess machines' ability to exhibit intelligent behaviour comparable to that of a human.

7. List capabilities needed by computers to pass Turing Test.

- natural language processing to enable it to communicate successfully in English.
- knowledge representation to store what it knows or learns;
- automated reasoning to use the stored information to answer questions and to draw new conclusions;
- machine learning to adapt to new circumstances and to detect and extrapolate patterns.

8. Define cognitive modelling.

Cognitive modeling involves creating computer models that simulate human cognitive processes to test theories about how the mind works. By comparing the behavior of these models to human behavior, researchers can determine if the models accurately represent human thought processes.

9. List any four foundations of AI.

1. Philosophy
2. Mathematics and statistics
3. Economics
4. Neuro science

10. List any four Applications of AI.

1. AI in Astronomy
2. AI in Healthcare
3. AI in Gaming
4. AI in Education

11. Define percept and percept sequence.

12. Define performance measure.

A performance measure embodies the criterion for success of an agent's behaviour. When an agent is placed down in an environment, it generates a sequence of actions according to the percepts it receives. This sequence of actions causes the environment to go through a sequence of states.

13. List any four state of the art AI applications.

1. Natural Language Processing (NLP) and Conversational AI
2. Computer Vision and Image Recognition
3. Autonomous Vehicles
4. Healthcare AI and Medical Diagnosis

14. Define agent. Give an example.

An agent is anything that can be viewed as perceiving its environment through sensors and SENSOR acting upon that environment through actuators.

Example: A human agent has eyes, ears, and other organs for sensors and hands, legs, mouth, and other

body parts for actuators. A robotic agent might have cameras and infrared range finders for sensors and various motors for actuators. A software agent receives keystrokes, file contents, and network packets as sensory inputs and acts on the environment by displaying on the screen, writing files, and sending network packets.

15. List various terminology associated with agent and environment.

Here are various terminologies associated with agents and environments:

Agent:

- Simple reflex agent
- Model-based reflex agent
- Goal-based agent
- Utility-based agent
- Learning agent

Environment:

- Fully observable environment
- Partially observable environment
- Deterministic environment

- Probabilistic environment
- Episodic environment
- Sequential environment
- Dynamic environment
- Static environment

Agent-Environment Interaction:

- Perception
- Action
- Sensor
- Actuator

16. Differentiate between agent function and agent program.

Agent Function:

- **Definition:** The agent function defines the mapping from percept histories to actions. In simpler terms, it specifies what action the agent should take in response to a sequence of percepts (observations or inputs).
- **Nature:** It is an abstract concept that describes the agent's behavior in terms of its input-output mapping without detailing how this mapping is implemented.
- **Example:** For a vacuum-cleaning agent, the agent function could be to "clean any dirty square it perceives".

Agent Program:

- **Definition:** The agent program is an actual implementation of the agent function. It is the concrete representation of the agent's behavior in code or a specific algorithm.
- **Nature:** It is the actual software or hardware that embodies the agent function. It takes inputs (percepts) and produces outputs (actions) as per the defined agent function.
- **Example:** For the vacuum-cleaning agent, the agent program could be a piece of software that uses sensors to detect dirty squares and actuators to move and clean those squares.

17. Define rational agents.

A rational agent is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome. An agent is something that acts. Computer agents are not mere programs, but they are expected to have the following attributes also:

1. Operating under autonomous control
2. Perceiving their environment
3. Persisting over a prolonged time period
4. Adapting to change

18. What is task environments? Example.

We must think about task environments which are essentially the "problems" to which rational agents are the "solutions."

Example: Specifying the task environment of a simple vacuum-cleaner agent:

Performance Measure:

- Maximizes cleanliness of the environment (percentage of clean cells).
- Minimizes time and energy used for cleaning.

Environment:

- A grid of cells with dirt and static obstacles.
- Bounded by walls.

Actuators:

- Movement in four directions (up, down, left, right).
- Vacuum to clean dirt in the current cell.

Sensors:

- Detects dirt in the current cell.
- Detects obstacles in adjacent cells.
- Detects boundaries of the environment.

19. Differentiate between omniscience and rationality.

Omniscience	Rationality
Definition: Complete and infinite knowledge about all aspects of the environment and outcomes of all possible actions.	Definition: The ability to make decisions that maximise performance based on available information and prior knowledge.
Knowledge Scope: Knows the actual outcome of every action.	Knowledge Scope: Limited to the agent's percepts and prior knowledge.
Feasibility: Impractical and impossible to achieve in reality.	Feasibility: Practical and achievable.
Nature: Idealised and theoretical.	Nature: Realistic and applied in real-world scenarios.

20. Write PEAS for an automated taxi driver.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe: fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Figure 2.4 PEAS description of the task environment for an automated taxi.

21. Write PEAS for a medical diagnosis system. (ref below table)

Agent Type: Medical diagnosis System

P(performance measure): healthy patient, minimum costs, lawsuits.

E(Environment): patient, hospital, staff.

A(Actuators): Display questions, tests, diagnoses, treatments, referrals.

S(Sensors): Keyboard entry of symptoms, findings, patient's answer.

22. Write PEAS for a satellite image analysis system.. (ref below table)

23. Write PEAS for a part-picking robot. . (ref below table)

24. Write PEAS for a refinery controller. . (ref below table)

25. Write PEAS for an Interactive English tutor. . (ref below table)

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, minimize costs, lawsuits	Patient, hospital, staff	Display questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display categorization of scene	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Maximize purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Maximize student's score on test	Set of students, testing agency	Display exercises, suggestions, corrections	Keyboard entry

26. Write PEAS for a Vacuum cleaner agent.

PEAS (Performance measure, Environment, Actuators, Sensors) description for a vacuum cleaner agent:

- Performance measure: Cleanliness of the floor, efficient use of power, time taken to clean, minimal noise.
- Environment: Varied floor types (carpet, tile, hardwood), obstacles (furniture, walls), different room layouts.
- Actuators: Wheels for movement, vacuum suction, brushes, beater bar, dirt container.
- Sensors: Dirt sensors, cliff sensors, bump sensors, wheel encoders, cameras or infrared sensors for navigation.

27. What are fully observable and partially observable Environments? Give an example.

*Fully observable Environment

If an agent's sensors give it access to the complete state of the environment at each point in time, then we say that the task environment is fully observable.

A task environment is effectively fully observable if the sensors detect all aspects that are relevant to the choice of action; relevance, in turn, depends on the performance measure.

Fully observable environments are convenient because the agent need not maintain any internal state to keep track of the world.

Example: chess game-the entire board is visible to both players

***Partially observable Environment**

An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data

for example, a vacuum agent with only a local dirt sensor cannot tell whether there is dirt in other squares, and an automated taxi cannot see what other drivers are thinking.

28. What are Deterministic and stochastic Environments? Give an example.

1. Deterministic Environment:

- In a deterministic environment, outcomes are entirely predictable given initial conditions and rules governing the system. There is no randomness or uncertainty involved.

- Example: A pendulum swinging under ideal conditions (ignoring air resistance and other external factors) follows deterministic laws of physics. If you know the initial position and velocity, you can predict exactly where the pendulum will be at any future time.

2. Stochastic Environment:

- In a stochastic environment, outcomes are influenced by randomness or probability. Even with known initial conditions, there is variability in the outcomes due to random factors or events.

- Example: Stock market fluctuations are a classic example of a stochastic environment. While economic principles and market trends influence stock prices, they are also affected by unpredictable events like economic crises, geopolitical events, or natural disasters. These factors introduce randomness into the outcomes.

29. What are Episodic and sequential Environments? Give an example.

Nested quantifiers are expressions in formal logic where one quantifier (such as $\forall$ or $\exists$) is contained within the scope of another quantifier. Here's an example:

Consider the domain of all integers, denoted by $\mathbb{Z}$.

Example: $\exists x \forall y (x + y = y)$

In this example:

$\exists x$ denotes "there exists an integer x",

$\forall y$ denotes "for all integers y".

30. What are Static and dynamic Environments? Give an example.

static environment, the conditions remain unchanged while an agent is deliberating. This means the agent can take its time to decide on an action without worrying about changes in

the environment or the passage of time. An example of a static environment is a crossword puzzle. Once the puzzle is presented, it remains the same until the agent (the person solving it) makes changes by filling in the answers.

dynamic environment changes while the agent is deliberating. The agent must continuously monitor the environment and quickly decide on actions to adapt to these changes. An example of a dynamic environment is taxi driving. The positions and actions of other vehicles, pedestrians, and traffic signals constantly change, requiring the driving algorithm to make rapid, real-time decisions to navigate safely.

31. What are Discrete and continuous Environments? Give an example.

Discrete vs. continuous: The discrete/continuous distinction can be applied to the state of the environment, to the way time is handled, and to the percepts and actions of the agent. For example, a discrete-state environment such as a chess game has a finite number of distinct states. Chess also has a discrete set of percepts and actions. Taxi driving is a continuous- state and continuous-time problem: the speed and location of the taxi and of the other vehicles sweep through a range of continuous values and do so smoothly over time. Taxi-driving actions are also continuous (steering angles, etc.).

32. What are Single agent and multiagent? Give an example.

33. Distinguish between fully and partially observable environments.

Aspect	Fully Observable Environment	Partially Observable Environment
Definition	The agent's sensors give it access to the complete state of the environment at each point in time.	The agent's sensors do not have access to the complete state of the environment at each point in time.
Effectiveness	Effectively fully observable if sensors detect all aspects relevant to the choice of action. Relevance depends on the performance measure.	Sensors are noisy, inaccurate, or parts of the state are missing from the sensor data.
Convenience	Convenient because the agent need not maintain any internal state to keep track of the world.	Requires the agent to maintain some internal state to infer missing information or deal with uncertainty.
Example Scenario	A scenario where all relevant information is available to the agent, e.g., a chess game where all pieces are visible.	A vacuum agent with a local dirt sensor cannot detect dirt in other squares; an automated taxi cannot perceive other drivers' intentions.

34. List the limitations of table driven agents.

TABLE-DRIVEN-AGENT does do what we want: it implements the desired agent function. The key challenge for AI is to find out how to write programs that, to the extent possible,

produce rational behavior from a small amount of code rather than from a large number of table entries.

35. List the four basic kinds of agent programs.

Simple reflex agents

Model-based reflex agents

Goal-based agents

Utility-based agents

36. What are the limitations of simple reflex agent?

37. What is a utility function?

An agent's preferences between world states are captured by a utility function, which assigns a single number to express the desirability of a state. Utilities are combined with outcome probabilities of actions to give an expected utility for each action.

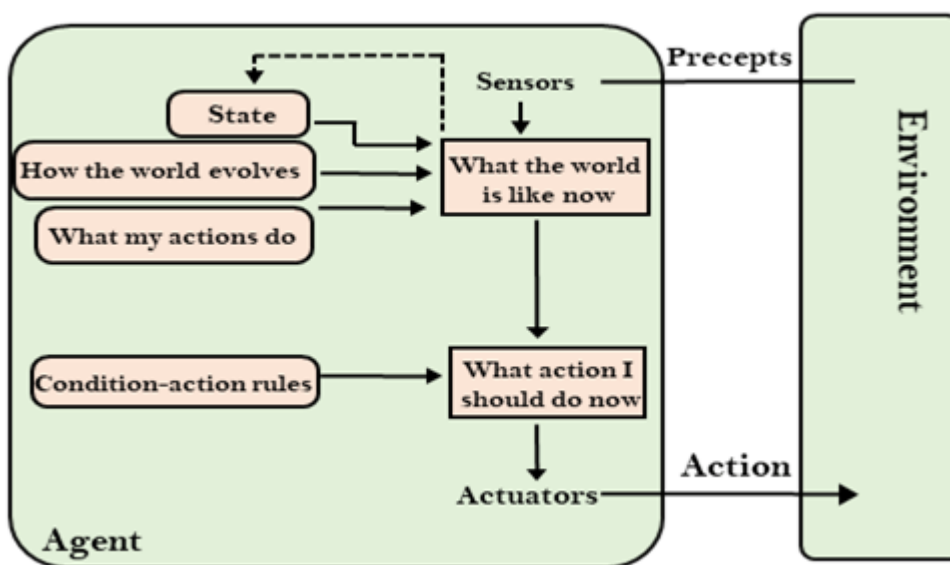
38. Distinguish between goal based and utility based agents.

Aspect	Goal-Based Agents	Utility-Based Agents
Definition	Aim to achieve a specific goal	Aim to maximise overall utility
Decision Process	Plan and execute actions to reach the goal	Evaluate and choose actions based on utility
Behaviour	Focus on reaching a specific end state	Consider multiple factors and trade-offs
Flexibility	Less adaptable to changing circumstances	More flexible and adaptive
Example	Robot navigating a maze to reach the exit	Self-driving car balancing safety and efficiency

39. What are Model-based reflex agents?

- A model-based agent has two important factors:
 - **Model:** It is knowledge about "how things happen in the world," so it is called a Model-based agent.

- **Internal State:** It is a representation of the current state based on percept history.
- These agents have the model, "which is knowledge of the world" and based on the model they perform actions.
- Updating the agent state requires information about:
 1. How the world evolves
 2. How the agent's action affects the world.



40. List the four conceptual components of a learning agent.

- A learning agent has mainly four conceptual components, which are:
 1. **Learning element:** It is responsible for making improvements by learning from environment
 2. **Critic:** Learning element takes feedback from critic which describes that how well the agent is doing with respect to a fixed performance standard.
 3. **Performance element:** It is responsible for selecting external action
 4. **Problem generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.

Long Answer Questions (3,4,5,6 Marks)

1. Write a note on “Acting humanly, the Turing test approach”.

Acting humanly: The Turing Test approach

- o Test proposed by Alan Turing in 1950
- o The computer is asked questions by a human interrogator.

The computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or not. Programming a computer to pass, the computer need to possess the following capabilities:

- Natural language processing to enable it to communicate successfully in English.
- Knowledge representation to store what it knows or hears
- Automated reasoning to use the stored information to answer questions and to draw new conclusions
- Machine learning to adapt to new circumstances and to detect and extrapolate patterns

To pass the complete Turing Test, the computer will need

- Computer vision to perceive the objects
- Robotics to manipulate objects and move about.

2. Explain “Thinking humanly: The cognitive modeling approach”

Thinking humanly: The cognitive modeling approach

We need to get inside actual working of the human mind:

- through introspection – trying to capture our own thoughts as they go by;
- through psychological experiments

Allen Newell and Herbert Simon, who developed GPS, the “General Problem Solver” tried to trace the reasoning steps to traces of human subjects solving the same problems. The interdisciplinary field of cognitive science brings together computer models from AI and experimental techniques from psychology to try to construct precise and testable theories of the workings of the human mind

3. Explain “Acting rationally: The rational agent approach”

An agent is something that acts. Computer agents are not mere programs, but they are expected to have the following attributes also :

- operating under autonomous control,
- perceiving their environment,

- persisting over a prolonged time period,
- adapting to change.

A rational agent is one that acts so as to achieve the best outcome.

4. Explain “Thinking rationally: The laws of thought approach”

The Greek philosopher Aristotle was one of the first to attempt to codify “right thinking”, that is irrefutable reasoning processes. His syllogism provided patterns for argument structures that always yielded correct conclusions when given correct premises—for example, “Socrates is a man; all men are mortal; therefore Socrates is mortal.”. These laws of thought were supposed to govern the operation of the mind; their study initiated a field called logic.

5. Briefly explain any five foundations of AI.

Economics

- How should we make decisions so as to maximize payoff?
- How should we do this when the payoff may be far in the future?
- ❖ The science of economics got its start in 1776, when Scottish philosopher Adam Smith treat it as a science, using the idea that economies can be thought of as consisting of individual agents maximizing their own economic well being.
- ❖ Decision theory, which combines probability theory with utility theory, provides a formal and complete framework for decisions (economic or otherwise) made under uncertainty—that is, in cases where probabilistic descriptions appropriately capture the decision maker’s environment.

Psychology:

- How do humans and animals think and act?
- ❖ The Behaviorism movement, led by John Watson(1878-1958).
- ❖ Behaviorists insisted on studying only objective measures of the percepts(stimulus) given to an animal and its resulting actions(or response).
- ❖ Behaviorism discovered a lot about rats and pigeons but had less success at understanding human.
- ❖ Cognitive psychology, views the brain as an information processing device.
- ❖ Common view among psychologist that a cognitive theory should be like a computer program.(Anderson 1980) i.e. It should describe a detailed information processing mechanism whereby some cognitive function might be implemented.

Neuroscience:

- How do brain process information?
 - ❖ Neuroscience is the study of the nervous system, particularly the brain.
 - ❖ Aristotle wrote, "Of all the animals, man has the largest brain in proportion to his size.

- ❖ Nicolas Rashevsky (1936, 1938) was the first to apply mathematical models to the study of the nervous system.
- ❖ The measurement of intact brain activity began in 1929 with the invention by Hans Berger of the electroencephalograph (EEG).
- ❖ The recent development of functional magnetic resonance imaging (fMRI) is giving neuroscientists detailed images of brain activity, enabling measurements that correspond in interesting ways to ongoing cognitive processes

Control theory and cybernetics:

- How can artifacts operate under their own control?
- ❖ Ktesibios of Alexandria (c. 250 B.C.) built the first self-controlling machine: a water clock with a regulator that maintained a constant flow rate.
- ❖ This invention changed the definition of what an artifact could do.
- ❖ Modern control theory, especially the branch known as stochastic optimal control, has as its goal the design of systems that maximize an objective function over time.
- ❖ This roughly matches our view of AI: designing systems that behave optimally.
- ❖ The tools of logical inference and computation allowed AI researchers to consider problems such as language, vision, and planning that fell completely outside the control theorist's purview.

Computer engineering:

- How can we build an efficient computer?
- ❖ For artificial intelligence to succeed, we need two things: intelligence and an artifact.
- ❖ The computer has been the artifact(object) of choice.
- ❖ The first operational computer was the electromechanical Heath Robinson, built in 1940 by Alan Turing's team for a single purpose: deciphering German messages.
- ❖ The first operational programmable computer was the Z-3, the invention of KonradZuse in Germany in 1941
- ❖ The first electronic computer, the ABC, was assembled by John Atanasoff and his student Clifford Berry between 1940 and 1942 at Iowa State University.
- ❖ The first programmable machine was a loom, devised in 1805 by Joseph Marie Jacquard (1752- 1834) that used punched cards to store instructions for the pattern to be woven.

6. Explain any six AI applications.

1. AI in Astronomy

- Artificial Intelligence can be very useful to solve complex universe problems. AI technology can be helpful for understanding the universe such as how it works, origin, etc.

2. AI in Healthcare

- In the last, five to ten years, AI becoming more advantageous for the healthcare industry and going to have a significant impact on this industry.
- Healthcare Industries are applying AI to make a better and faster diagnosis than humans. AI can help doctors with diagnoses and can inform when patients are worsening so that medical help can reach to the patient before hospitalization.

3. AI in Gaming

- AI can be used for gaming purpose. The AI machines can play strategic games like chess, where the machine needs to think of a large number of possible places.

4. AI in Robotics:

- Artificial Intelligence has a remarkable role in Robotics. Usually, general robots are programmed so that they can perform some repetitive task, but with the help of AI, we can create intelligent robots which can perform tasks with their own experiences without pre-programmed.
- Humanoid Robots are best examples for AI in robotics, recently the intelligent Humanoid robots named as Erica and Sophia has been developed which can talk and behave like humans.

5. AI in Entertainment

- We are currently using some AI based applications in our daily life with some entertainment services such as Netflix or Amazon. With the help of ML/AI algorithms, these services show recommendations for programs or shows.

6. AI in Agriculture

- Agriculture is an area which requires various resources, labor, money, and time for best result. Nowadays agriculture is becoming digital, and AI is emerging in this field. Agriculture is applying agriculture robotics, solid and crop monitoring, predictive analysis. AI in agriculture can be helpful for farmers.

7. Explain any six state of the art AI applications.

Autonomous control: The ALVINN computer vision system was trained to steer a car to keep it following a lane. It was placed in CMU's NAVLAB computer-controlled minivan and used to navigate across the United States.

Game playing: IBM's Deep Blue became the first computer program to defeat the world champion in a chess match when it bested Garry Kasparov by a score of 3.5 to 2.5 in an exhibition match (Goodman and Keene, 1997).

Diagnosis: Medical diagnosis programs based on probabilistic analysis have been able to perform at the level of an expert physician in several areas of medicine.

Autonomous planning and scheduling: A hundred million miles from Earth, NASA's Remote Agent program became the first on-board autonomous planning program to control the scheduling of operations for a spacecraft (Jonsson et al., 2000). Remote Agent generated

plans from high-level goals specified from the ground, and it monitored the operation of the spacecraft as the plans were executed-detecting, diagnosing, and recovering from problems as they occurred.

Robotics: Many surgeons now use robot assistants in microsurgery. HipNav (DiGioia et al., 1996) is a system that uses computer vision techniques to create a three-dimensional model of a patient's internal anatomy and then uses robotic control to guide the insertion of a hip replacement prosthesis.

Language understanding and problem solving: PROVERB (Littman et al., 1999) is a computer program that solves crossword puzzles better than most humans, using constraints on possible word fillers, a large database of past puzzles, and a variety of information sources including dictionaries and online databases such as a list of movies and the actors that appear in them.

8. Explain History of AI.

The gestation of artificial intelligence (1943-1955) There were a number of early examples of work that can be characterized as AI, but it was Alan Turing who first articulated a complete vision of AI in his 1950 article "Computing Machinery and Intelligence." Therein, he introduced the Turing test, machine learning, genetic algorithms, and reinforcement learning.

The birth of artificial intelligence (1956) McCarthy convinced Minsky, Claude Shannon, and Nathaniel Rochester to help him bring together U.S. researchers interested in automata theory, neural nets, and the study of intelligence. They organized a two-month workshop at Dartmouth in the summer of 1956. Perhaps the longest-lasting thing to come out of the workshop was an agreement to adopt McCarthy's new name for the field – Artificial Intelligence

Early enthusiasm, great expectations (1952-1969) The early years of AI were full of successes-in a limited way.

General Problem Solver (GPS) was a computer program created in 1957 by Herbert Simon and Allen Newell to build a universal problem solver machine. The order in which the program considered subgoals and possible actions was similar to that in which humans approached the same problems. Thus, GPS was probably the first program to embody the "thinking humanly" approach. At IBM, Nathaniel Rochester and his colleagues produced some of the first AI programs. Herbert Gelernter (1959) constructed the Geometry Theorem Prover, which was able to prove theorems that many students of mathematics would find quite tricky. Lisp was invented by John McCarthy in 1958 while he was at the Massachusetts Institute of Technology (MIT). In 1963, McCarthy started the AI lab at Stanford. Tom Evans's ANALOGY program (1968) solved geometric analogy problems that appear in IQ tests

A dose of reality (1966-1973) From the beginning, AI researchers were not shy about making predictions of their coming successes. The following statement by Herbert Simon in 1957 is often quoted: "It is not my aim to surprise or shock you-but the simplest way I can summarize is to say that there are now in the world machines that think, that learn and that create. Moreover, their ability to do these things is going to increase rapidly until-in a visible

future-the range of problems they can handle will be coextensive with the range to which the human mind has been applied.

Knowledge-based systems: The key to power? (1969-1979) Dendral was an influential pioneer project in artificial intelligence (AI) of the 1960s, and the computer software expert system that it produced. Its primary aim was to help organic chemists in identifying unknown organic molecules, by analyzing their mass spectra and using knowledge of chemistry. It was done at Stanford University by Edward Feigenbaum, Bruce Buchanan, Joshua Lederberg, and Carl Djerassi.

AI becomes an industry (1980-present) In 1981, the Japanese announced the "Fifth Generation" project, a 10-year plan to build intelligent computers running Prolog. Overall, the AI industry boomed from a few million dollars in 1980 to billions of dollars in 1988.

The return of neural networks (1986-present) Psychologists including David Rumelhart and Geoff Hinton continued the study of neural-net models of memory.

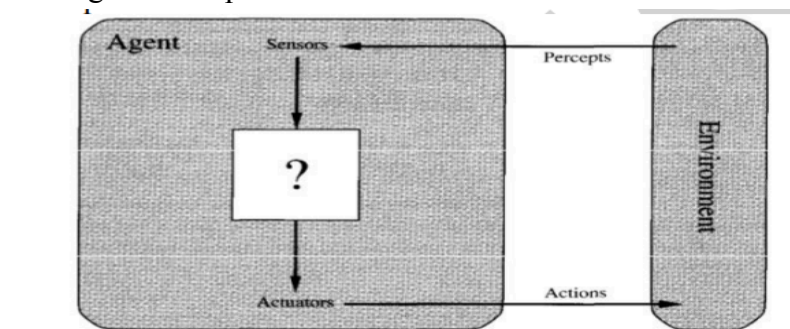
AI becomes a science (1987-present) In recent years, approaches based on hidden Markov models (HMMs) have come to dominate the area. Speech technology and the related field of handwritten character recognition are already making the transition to widespread industrial and consumer applications. The Bayesian network formalism was invented to allow efficient representation of, and rigorous reasoning with, uncertain knowledge.

The emergence of intelligent agents (1995-present) One of the most important environments for intelligent agents is the Internet.

9. Explain the concept of agents and environments with an example.

An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators. This simple idea is illustrated in Figure 1.2

- A human agent has eyes, ears, and other organs for sensors and hands, legs, mouth, and other body parts for actuators.
- A robotic agent might have cameras and infrared range finders for sensors and various motors for actuators.
- A software agent receives keystrokes, file contents, and network packets as sensory inputs and acts on the environment by displaying on the screen, writing files, and sending network packets



10. Write a note on rationality and rational agent.

Rational Agent :A rational agent is one that does the right thing-conceptually speaking, every entry in the table for the agent function is filled out correctly. Obviously, doing the right thing is better than doing the wrong thing. The right action is the one that will cause the agent to be most successful.

Rationality What is rational at any given time depends on four things:

- The performance measure that defines the criterion of success.
- The agent's prior knowledge of the environment.
- The actions that the agent can perform.
- The agent's percept sequence to date.

This leads to a definition of a rational agent: For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

11.How do you specify a task environment using PEAS description? Explain with an example.

We must think about task environments, which are essentially the "problems" to which rational agents are the "solutions." Specifying the task environment The rationality of the simple vacuum-cleaner agent, needs specification of

- the performance measure
- the environment
- the agent's actuators and sensors.

PEAS :All these are grouped together under the heading of the task environment. We call this the PEAS (Performance, Environment, Actuators, Sensors) description. In designing an agent, the first step must always be to specify the task environment as fully as possible

Agent Type	Performance Measure	Environments	Actuators	Sensors
Taxi driver	Safe: fast, legal, comfortable trip, maximize profits	Roads,other traffic,pedestrians, customers	Steering,accelerator, brake, Signal,horn,display	Cameras,sonar, Speedometer,GPS, Odometer,engine sensors,keyboards, accelerometer

Figure 1.5 PEAS description of the task environment for an automated taxi.

12.Explain rationality with example.

Rationality What is rational at any given time depends on four things:

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13.Explain the various categories of environments with example.

Fully observable vs. partially observable. If an agent's sensors give it access to the complete state of the environment at each point in time, then we say that the task environment is fully observable. A task environment is effectively fully observable if the sensors detect all aspects that are relevant to the choice of action; An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data.

Deterministic vs. stochastic. If the next state of the environment is completely determined by the current state and the action executed by the agent, then we say the environment is deterministic; otherwise, it is stochastic.

Episodic vs. sequential In an episodic task environment, the agent's experience is divided into atomic episodes. Each episode consists of the agent perceiving and then performing a single action. Crucially, the next episode does not depend on the actions taken in previous episodes. For example, an agent that has to spot defective parts on an assembly line bases each decision on the current part, regardless of previous decisions; In sequential environments, on the other hand, the current decision could affect all future decisions. Chess and taxi driving are sequential:

Discrete vs. continuous. The discrete/continuous distinction can be applied to the state of the environment, to the way time is handled, and to the percepts and actions of the agent. For example, a discrete-state environment such as a chess game has a finite number of distinct states. Chess also has a discrete set of percepts and actions. Taxi driving is a continuous-state and continuous-time problem: the speed and location of the taxi and of the other vehicles sweep through a range of continuous values and do so smoothly over time. Taxi-driving actions are also continuous (steering angles, etc.).

Single agent vs. multiagent. An agent solving a crossword puzzle by itself is clearly in a single-agent environment, whereas an agent playing chess is in a two-agent environment. As one might expect, the hardest case is par

14.Explain Agent programs with example.

The agent programs all have the same skeleton: they take the current percept as input from the sensors and return an action to the actuators. The agent program which takes the current percept as input, and the agent function, which takes the entire percept history. The agent program takes just the current percept as input because nothing more is available from the environment; if the agent's actions depend on the entire percept sequence, the agent will have to remember the percepts.

Function TABLE-DRIVEN_AGENT(*percept*) **returns** an action

static: *percepts*, a sequence initially empty
 table, a table of actions, indexed by percept sequence

append *percept* to the end of *percepts*
action ← LOOKUP(*percepts*, *table*)
return *action*

Figure 1.8 The TABLE-DRIVEN-AGENT program is invoked for each new percept and returns an action each time.

15. Explain simple reflex agent with a neat diagram.

The simplest kind of agent is the simple reflex agent. These agents select actions on the basis of the current percept, ignoring the rest of the percept history. For example, the vacuum agent whose agent function is a simple reflex agent, because its decision is based only on the current location and on whether that contains dirt.

- Select action on the basis of only the current percept. E.g. the vacuum-agent
- Large reduction in possible percept/action situations(next page).
- Implemented through condition-action rules If dirty then suck

A Simple Reflex Agent: Schema

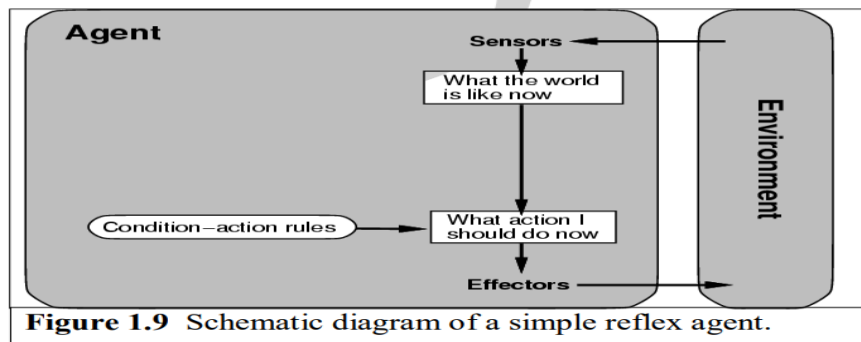


Figure 1.9 Schematic diagram of a simple reflex agent.

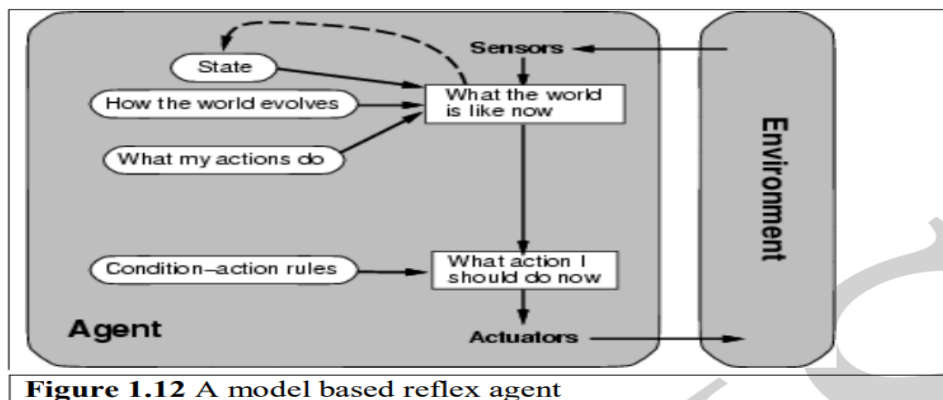
Characteristics

- Only works if the environment is fully observable.
- Lacking history, easily get stuck in infinite loops
- One solution is to randomize actions

16. Briefly explain model based reflex agent with a diagram.

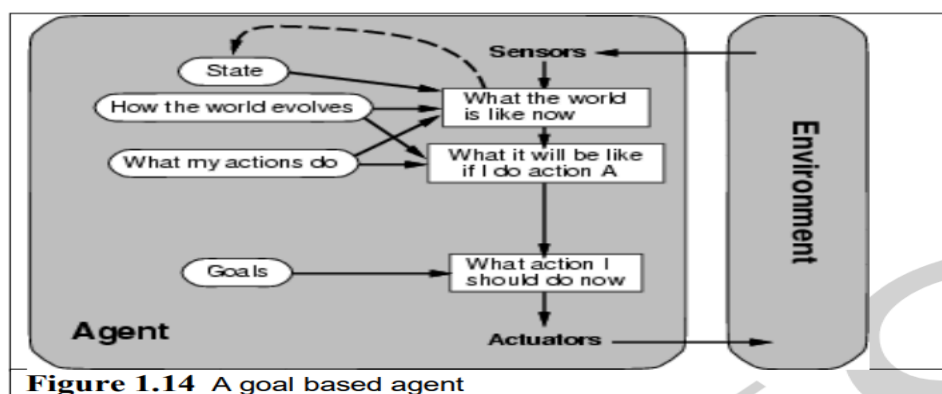
The most effective way to handle partial observability is for the agent to keep track of the part of the world it can't see now. That is, the agent should maintain some sort of internal state that depends on the percept history and thereby reflects at least some of the unobserved aspects of the current state. Updating this internal state information as time goes by requires two kinds of knowledge to be encoded in the agent program. First, we need some information about how the world evolves independently of the agent-for example, that an overtaking car

generally will be closer behind than it was a moment ago. Second, we need some information about how the agent's own actions affect the world—for example, that when the agent turns the steering wheel clockwise, the car turns to the right or that after driving for five minutes northbound on the freeway one is usually about five miles north of where one was five minutes ago. This knowledge about "how the world working - whether implemented in simple Boolean circuits or in complete scientific theories—is called a model of the world. An agent that uses such a MODEL-BASED model is called a model-based agent.



17. Briefly explain goal based reflex agent with a diagram

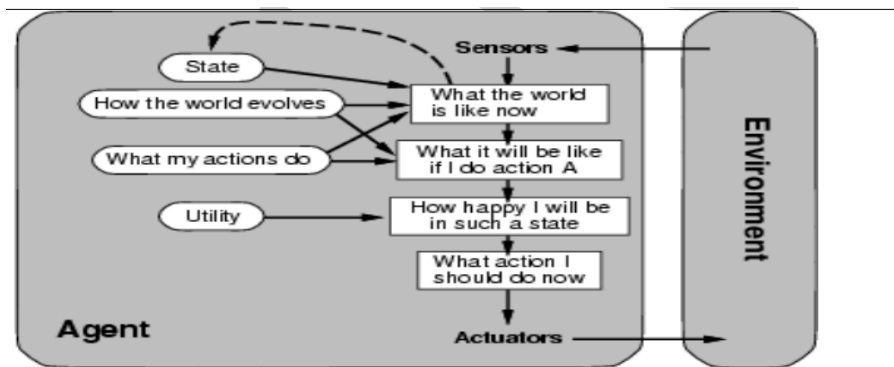
Knowing about the current state of the environment is not always enough to decide what to do. For example, at a road junction, the taxi can turn left, turn right, or go straight on. The correct decision depends on where the taxi is trying to get to. In other words, as well as a current state description, the agent needs some sort of goal information that describes situations that are desirable—for example, being at the passenger's destination. The agent program can combine this with information about the results of possible actions (the same information as was used to update internal state in the reflex agent) in order to choose actions that achieve the goal. Figure 1.13 shows the goal-based agent's structure.



18. Explain utility based agent with a diagram.

Goals alone are not really enough to generate high-quality behavior in most environments. For example, there are many action sequences that will get the taxi to its destination (thereby

achieving the goal) but some are quicker, safer, more reliable, or cheaper than others. Goals just provide a crude binary distinction between "happy" and "unhappy" states, whereas a more general performance measure should allow a comparison of different world states according to exactly how happy they would make the agent if they could be achieved. Because "happy" does not sound very scientific, the customary terminology is to say that if one world state is preferred to another, then it has higher utility for the agent.



19. Explain learning agent with a diagram

All agents can improve their performance through learning. A learning agent can be divided into four conceptual components, as shown in Figure 1.15. The most important distinction is between the learning element, which is responsible for making improvements, and the performance element, which is responsible for selecting external actions. The performance element is what we have previously considered to be the entire agent: it takes in percepts and decides on actions. The learning element uses feedback from the critic on how the agent is doing and determines how the performance element should be modified to do better in the future. The last component of the learning agent is the problem generator. It is responsible for suggesting actions that will lead to new and informative experiences. But if the agent is willing to explore a little, it might discover much better actions for the long run. The problem generator's job is to suggest these exploratory actions. This is what scientists do when they carry out experiments.

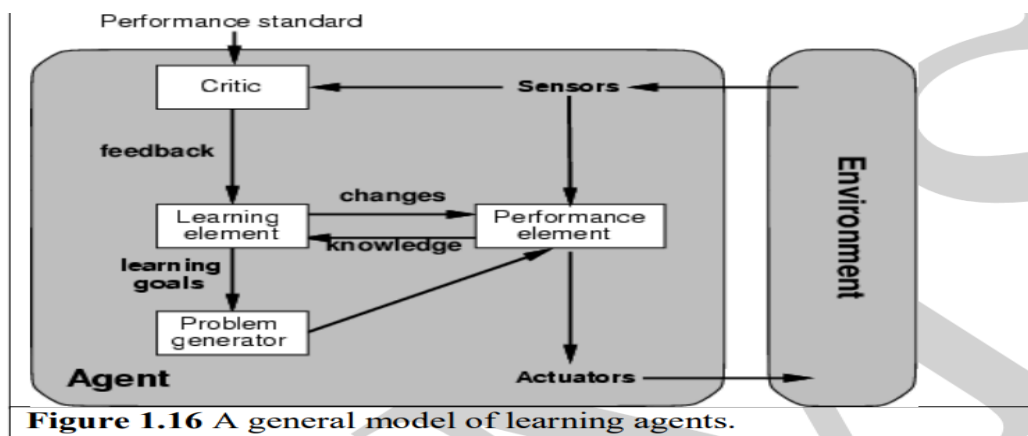


Figure 1.16 A general model of learning agents.